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Mast Cells in Host Defense Against Cutaneous *Staphylococcus aureus* Infection

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ABSTRACT

Staphylococcus aureus is an opportunistic pathogen implicated in skin and soft-tissue infections and chronic inflammatory skin diseases, such as atopic dermatitis. These infections can range from mild to potentially life-threatening systemic illnesses progressing to bacteremia, endocarditis, and sepsis. Mast cells, traditionally recognized for their role in allergies, are highly abundant within the skin and are gaining recognition for their contribution to bacterial defense. Here, we discuss the role of mast cells in three models of *S. aureus* skin infection according to skin depth: epicutaneous sensitization, intradermal injections, and subcutaneous injections. During *S. aureus* skin infection, mast cells become activated and accumulate in infected skin, recognize bacterial toxins, and modulate the activity of other immune cells, including neutrophils and dendritic cells. We discuss areas of research that should be the target of future studies, such as neuroimmunology, infected excisional wounds, and risk of systemic illness. Our review aims to bring attention to the host–pathogen interaction between mast cells and *S. aureus* in the skin, both to encourage deeper investigation and inform the development of immunomodulatory-based therapeutics.

1 | Introduction

Skin and soft-tissue infections (SSTIs) continue to pose a major threat to public health despite advances in modern medicine. In 2012, the direct healthcare costs of SSTIs in the United States more than tripled to $15.0 billion from $4.8 billion in 2000 [1]. Between 2010 and 2020 in the United States, the overall incidence of SSTIs was 77.5 per 1000 person-years of observation, highlighting the severity of the situation [2]. *Staphylococcus aureus* is the most common pathogen isolated from SSTI cultures in the United States, causing infections ranging from mild to life-threatening systemic illnesses [3]. Treatment is complicated by the antibiotic susceptibility profiles of SSTI isolates, with a large proportion classified as methicillin-resistant *S. aureus* (MRSA) [4]. In addition to SSTIs, *S. aureus* is implicated in a variety of chronic inflammatory skin conditions that severely impact quality of life, including atopic dermatitis (AD) [5], Netherton syndrome [6], psoriasis [7], and bullous pemphigoid [8].

The skin is the largest organ in the human body and is vital to protecting the host from exogenous insults. The skin is

Abbreviations: AD, Atopic dermatitis; Agr, Accessory gene regulator; AIP, Autoinducing peptide; BMMCs, Bone marrow-derived mast cells; CBMC, Cord blood-derived mast cells; Cpa3, Carboxypeptidase A3 promoter; CRAMP, Cathelicidin-related antimicrobial peptide; CSP-1, Competence stimulating peptide-1; CTMC, Connective tissue mast cell; DAMPS, Damage-associated molecular patterns; DEG, Differentially expressed gene; ESST, Ear skin and soft tissue; EV, Extracellular vesicle; GMP, Granulocyte monocyte progenitor; HMC-1, Human mast cell line-1; HSC, Hematopoietic stem cell; IFN, Interferon; IFNAR, Interferon α receptor; IgE, Immunoglobulin E; IL, Interleukin; LDH, Lactate dehydrogenase; LL-37, Cathelicidin; LPS, Lipopolysaccharide; LTC_4, Leukotriene C_4; MC_T, Tryptase$^+$ mast cell; MC_{TC}, Tryptase$^+$Chymase$^+$ mast cell; MMC, Mucosal mast cell; MOI, Multiplicities of infection; MRSA, Methicillin-resistant *Staphylococcus aureus*; OHM, ω-hydroxyemodin; OVA, Ovalbumin; PAMPs, Pathogen-associated molecular patterns; PGN, Peptidoglycan; PRRs, Pattern recognition receptors; PSM, Phenol-soluble modulin; SCF, Stem cell factor; SEB, Staphylococcal enterotoxin B; SP, Substance P; SSTI, Skin and soft-tissue infection; TLR, Toll-like receptor; TNF, Tumor necrosis factor.

equipped with a complex system of physical, chemical, microbial, and immune defenses. Among the numerous immune cells that populate the skin, mast cells have recently gained attention for their roles in bacterial defense [9]. While first thought to solely function in allergic reactions, emerging evidence has now underscored their significance in a broad range of infectious contexts, which have been the focus of numerous high-impact review papers [10–12].

The purpose of this review was to provide a comprehensive overview of studies investigating the host–pathogen interaction between mast cells and *S. aureus* in the skin. For an overarching understanding of the innate and adaptive immune responses against cutaneous *S. aureus*, we refer readers to reviews published by Krishna and Miller [13] and Liu et al. [14]. As relatively new cells in the field of host–pathogen immunity, we hope to shed light on their versatile functions in pathogen surveillance. In this review, we will go beyond the synthesis of literature and critically appraise studies, both to inform therapeutic design and emphasize areas which require further research. Our review will discuss three modes of infection which correspond to differences in skin depth: epicutaneous sensitization, intradermal injection, and subcutaneous injection.

2 | Mast Cells

Mast cells are tissue resident innate immune cells belonging to the hematopoietic lineage [15, 16]. In rodents, mast cells are subdivided into two types, connective tissue mast cells (CTMC) and mucosal mast cells (MMC) [17]. CTMCs and MMCs differ in numerous properties, such as anatomical location, size, granule contents, and receptor expression [18, 19]. Within secretory granules, the sulfated proteoglycans in CTMCs are primarily composed of heparin, while MMC proteoglycans contain chondroitin sulphate [20–22]. In terms of protease content, CTMCs often express the chymase mMCP-4, the elastase mMCP-5, tryptases mMCP-6 and 7, and carboxypeptidase A3 [23, 24]. While MMCs express the chymases and mMCP-1 and 2 [23–25]. However, mast cells can exhibit tissue-specific heterogeneity in protease expression [23, 24]. CTMCs primarily arise during embryogenesis from early-erythromyeloid progenitors that are gradually replaced by late-erythromyeloid progenitors derived from the embryonic yolk sac, with the exception of adipose tissue and the pleural cavity which retain early-erythromyeloid progenitors [19, 26]. MMC in the gut epithelium and respiratory mucosa are derived from fetal hematopoietic stem cells (HSCs) [26]. CTMC are long-lived and maintained locally through self-renewal, while MMC are inducible, transient, and replaced by bone marrow-derived mast cell progenitors [26–29]. In humans, these subsets correspond to mast cells containing the neutral proteases chymase and tryptase (MC_{TC}) versus mast cells that exclusively contain tryptase (MC_T) [30]. Recently, Tauber et al., revealed that mouse CTMC which express the G protein-coupled receptor Mrgprb2, are transcriptionally distinct from Mrgprb2^{negative} MMCs. Conversely, humans contain at least six mast cell states with heterogenous expression of MRGPRX2, the human ortholog of Mrgprb2 [19, 31].

Mast cell maturation and differentiation depend on stem cell factor (SCF, the ligand for c-kit) with the help of auxiliary cytokines and growth factors, including interleukin (IL)-3 and IL-6 [16, 32, 33]. Mammalian mast cells are round or elongated in morphology with non-segmented circular nuclei [16, 30, 34]. Mast cells contain electron-dense lipid bodies and cytoplasmic granules that stain metachromatically with toluidine blue [16]. Evidence suggests that mast cells originated more than 500 million years ago, prior to the emergence of chordates or adaptive immunity [35]. Mast cells have been conserved throughout evolution as no human has ever been reported without them, demonstrating their vital role in innate immunity [35].

Mast cells are best known for their role in type I hypersensitivity reactions, where they act as initiators of allergic inflammation [36, 37]. The canonical pathway of mast cell activation is mediated by immunoglobulin E (IgE) which binds to the high-affinity FcεRI receptor on the surface of mast cells [36]. When mast cells are sensitized with IgE and exposed to a specific antigen or allergen, the addition of cross-linking stimuli leads to degranulation within seconds to minutes [36]. Degranulation consists of the release of preformed pro-inflammatory mediators with immunoregulatory properties, such as acid hydrolases, histamine, proteoglycans, proteases, and tumor necrosis factor (TNF) [36, 38–41]. Mast cells subsequently release de novo-synthesized lipid mediators, including platelet-activating factor, leukotrienes, and prostaglandins [42, 43]. Finally, mast cells release a variety of cytokines and chemokines which must be transcribed and translated prior to release, such as IL-6 and CCL2 [37, 44]. Mast cells can also be induced to degranulate through IgE-independent also known as non-immunologic activation. For instance, McNeil et al. proved that mast cells degranulate through MRGPRX2 in humans and Mrgprb2 in mice, in response to cationic polybasic compounds such as antimicrobial peptides, neuropeptides, and several clinically used drugs [31, 45]. Intriguingly, neurogenic Mrgprb2 activation following incision injury was shown to mediate postoperative inflammatory pain and innate immune cell recruitment [46].

3 | Mast Cells in Bacterial Infections

Mast cells most commonly reside at junctures between the host and environment, such as the skin, lungs, small intestine, and urogenital tract [30, 47]. In the skin, mast cells are located in the subepidermal layers with the highest abundance in the upper papillary dermis [48, 49]. Mast cell numbers progressively decline as the skin gets deeper, with the subcutaneous layer containing up to one-tenth the number of mast cells as the upper papillary dermis [41]. Skin mast cells preferentially reside in close contact with hair follicles, blood vessels, and peripheral neurons, presumably due to their effector functions [48–52]. Due to their temporal advantage owing to their ability to rapidly release preformed mediators and spatial advantage, mast cells are believed to be one of the first leukocytes to respond to pathogens that endanger the host [10, 53–55]. Mast cell-deficient mice are more vulnerable to a multitude of bacterial infections, including *Pseudomonas aeruginosa*, *Listeria monocytogenes*, and group A streptococci, underscoring their importance in pathogen detection [56–58]. Mast cells are armed with a variety of surface and intracellular receptors that recognize pathogens, collectively referred to as pattern recognition receptors (PRRs) [47, 59]. PRRs recognize invariant molecules expressed by pathogens, called pathogen-associated molecular patterns (PAMPs) or molecules released from dying or infected host cells called damage-associated molecular patterns (DAMPs) [53, 59]. Recent research from our lab identified MRGPRX2 and Mrgprb2 as human and mouse mast cell receptors, respectively, for quorum sensing molecules produced by *Streptococcus pneumoniae* [9]. Competence stimulating peptide-1 (CSP-1) produced by *S. pneumoniae* was able to promote mast cell degranulation through MRGPRX2/b2. In a nasopharyngeal infection model, Mrgprb2 was shown to improve infection outcome by promoting TNF production which leads to leukocyte recruitment and bacterial clearance. Furthermore, we have identified a mechanism by which mast cells can detect bacterial quorum sensing molecules [9], establishing MRGPRX2/b2 as a key early sensor in bacterial infection.

4 | Direct Crosstalk Between Mast Cells and *S. aureus*

Mast cells respond to live *S. aureus* in vitro and are known to exhibit distinct responses to *S. aureus* secreted toxins or cell wall-anchored components (Table 1). In a coculture experiment, bone marrow-derived mast cells (BMMCs) inhibited the growth of extracellular *S. aureus* via mast cell extracellular trap formation and the release of potent antibacterial mediators during degranulation [60]. A proportion of *S. aureus* was found to be internalized within BMMCs and human mast cell line-1 (HMC-1) mast cells following infection [60, 61]. Interestingly, *S. aureus* internalization within mast cells was shown to be dependent on a reciprocal relationship. Mast cell granules promote the upregulation of virulence factors, including α-hemolysin, a cytolytic pore-forming toxin that induces tissue necrosis and immune cell filtration, as well as fibronectin-binding protein A, an adhesive protein responsible for mediating adherence to host cells [62–64]. In turn, α-hemolysin stimulates β1 integrin expression on mast cells, enabling *S. aureus* to bind to β1 integrin via fibronectin thereby facilitating translocation into the cell [62].

In one study, intracellular *S. aureus* persisted and remained viable 5 days postinfection, demonstrating that *S. aureus* may have evolved this host-evasion strategy to subvert mast cell killing [60]. Another study observed a decrease in the number of viable bacteria across a 24-h period [61]. This discrepancy likely reflects the multiplicities of infection (MOI) of 20:1 [60] *S. aureus* to mast cells versus 5:1 [61], respectively. It is plausible that a higher ratio of *S. aureus* relative to mast cells aids the bacteria in eluding the intracellular antibacterial responses. It remains to be determined which MOI is more relevant to in vivo infection models. *S. aureus* internalization within mast cells has been corroborated in human cord blood-derived mast cells (CBMC) [66], but has yet to be shown using terminally differentiated mast cells that resemble a

TABLE 1 | *Staphylococcus aureus* components reported to activate mast cells or modulate mast cell function.

***S. aureus* component**	**Outcome**	**Source(s)**
α-hemolysin	Mediates intracellular survival of *S. aureus* within bone marrow-derived mast cells (BMMCs) and human mast cell line-1 (HMC-1) mast cells by modulating β1 integrin expression	[60, 62]
	Induces dose-dependent degranulation in peritoneal mast cells	[65]
Peptidoglycan (PGN)	PGN induces IL-8 and TNF release from cord blood-derived mast cells (CBMCs)	[66]
	PGN induces TNF release in a partial TLR2-dependent manner, degranulation, and cytokine production (IL-4, IL-5, IL-6, IL-10, IL-13)	[67]
Protein A	Promotes degranulation and de novo synthesis of cysteinyl leukotriene C_4 (LTC_4) in human cardiac mast cells by binding to IgE V_H3+ bound on FcεRI	[68, 69]
Delta toxin	Delta-toxin induces biphasic mast cell degranulation in BMMCs and connective tissue mast cells (CTMCs) through an early Ca^{2+} independent phase and a late extracellular Ca^{2+} dependent phase	[70]
	δ-toxin promotes the release of tryptase and lactate dehydrogenase (LDH) from HMC-1 mast cells	[71]
Phenol-soluble modulin (PSM)-α 1, 2, and 3	PSM-α 1, 2, and 3 induce loss of mast cell viability as evidenced by LDH release	[71]
Superantigen-like 12	Induces degranulation and de novo synthesis of IL-6 and IL-13 by BMMCs	[72]
Staphylococcal enterotoxin B (SEB)	SEB inhibits the production, release, and transcription of IL-4 in HMC-1 cells, SEB also decreases TNF mRNA production in a biphasic manner	[73]

mature phenotype. In response to intracellular *S. aureus*, infected HMC-1 cells released interferon-α (IFN-α) through the cGAS-STING-TBK1 signaling pathway [61]. When the receptor for IFN-α, IFNAR, was blocked, the cells had an impaired ability to kill intracellular *S. aureus*, demonstrating that IFN-α acts in an autocrine manner to induce a bactericidal response within mast cells. Interestingly, both infected and neighboring cells exhibited a distinct IFN-I transcriptomic signature. This demonstrates that IFN-α acts on bystander cells in a paracrine manner and may prime these cells to respond in the event of bacterial invasion [61]. *S. aureus* can also induce mast cells to release pro-inflammatory cytokines, including vascular endothelial growth factor, IL-3, IL-8, IL-13, and TNF [60, 66, 74, 75].

It is important to note that caution should be taken when interpreting results using in vitro differentiated or immortalized cell lines due to their functional and phenotypic differences from mature tissue mast cells. BMMCs cultured with IL-3 differ significantly from CTMCs in their maturity, granule contents, receptor expression, and signaling programs [76]. For instance, BMMCs express markedly lower levels of the mast cell-specific proteases Mcpt4, Mcpt5, and Mcpt6, as well as over 100-fold less Mrgprb2 compared to peritoneal mast cells [76]. HMC-1 cells display only slight expression of FcεRI and reduced release of mast cell mediators during degranulation, such as histamine and tryptase [77–79]. Another issue with these in vitro experiments is their inability to mimic the complex microenvironment present within the skin, including potential indirect effects on mast cell function. They also exclude the potential influence of other cell types, growth factors, or cytokines on mast cell phenotype.

5 | Mast Cell Deficiency Models

Our current knowledge of mast cell biology has largely stemmed from the use of mast cell deficiency models in rodents. Traditional models contain spontaneous genetic mutations in the *c-kit* proto-oncogene allelic with the white spotting (*W*) locus [80–82]. *c-kit* encodes the transmembrane receptor tyrosine kinase kit (CD117), the receptor for SCF, which is important for mast cell survival and maturation [80, 81]. c-kit is also expressed on a variety of other cell types, such as HSCs [83], germ cells [84], and melanocytes [85]. WBB6F$_1$-Kit$^{W/W\text{-}v}$ mice contain a null *W* allele and a dominant W^v allele with a point mutation in *c-kit*, impairing the kinase activity of the receptor [81]. C57BL/6J-*Kit*$^{W\text{-}sh/W\text{-}sh}$ mice arise from an inversion mutation upstream of the *c-kit* transcription start site on chromosome 5 [80, 86]. As previously reviewed by Grimbaldeston et al. [87], both *Kit*$^{W/W\text{-}v}$ and *Kit*$^{W\text{-}sh/W\text{-}sh}$ mice exhibit a variety of phenotypic abnormalities which may confound interpretation of results. For instance, *Kit*$^{W\text{-}sh/W\text{-}sh}$ mice exhibit circulating neutrophilia while *Kit*$^{W/W\text{-}v}$ mice are mildly neutropenic [80]. This is problematic given the role that neutrophils are known to play during skin infection [13]. An effort has been made to verify mast cell dependency in these models through reconstitution with in vitro differentiated mast cells [88]. However, these experiments do not exclude the possibility that the observed effects are due to reduced Kit-signaling in other cell types, rather than mast cell deficiency [88]. There is also evidence that skin mast cells do not reach wildtype (WT) levels following adoptive transfer of donor mast cells [26, 89, 90].

There has been a recent shift in the field to the use of *c-kit*-independent models of mast cell depletion, such as *Cpa3-Cre*; *Mcl-1*$^{fl/fl}$. *Cpa3-Cre*; *Mcl-1*$^{fl/fl}$ mice are produced when mice expressing Cre recombinase under the control of the carboxypeptidase A3 promoter (*Cpa3*) are crossed with mice that have floxed *Mcl-1* alleles [91]. Mcl-1 is an antiapoptotic factor and when removed by Cre recombinase, results in marked mast cell depletion through apoptosis [92, 93]. However, since basophils also express *Cpa3*, this model results in significant depletion of basophils in the spleen, bone marrow, and blood [91]. Similarly, *Mcpt5-Cre* mice use a Cre-loxP system to conditionally deplete connective tissue mast cells [94, 95]. Although no model is perfect, using Kit-independent models to verify results originally obtained from Kit models provides increased certainty that experimental findings truly reflect mast cell involvement as opposed to artifacts of the model.

6 | Mast Cells in *S. aureus* Epicutaneous Sensitization Infection Models

The most extensively studied inflammatory skin disease involving *S. aureus* is AD. AD is a highly pruritic, relapsing chronic inflammatory skin disease that impacts up to 20% of children and 10% of adults in industrialized countries [96–98]. Many individuals with AD experience cutaneous microbial dysbiosis, primarily in the neck, antecubital, and popliteal fossae regions of the skin [5, 99]. 16S rRNA sequencing has revealed an enrichment of the *Staphylococcus* genus, with *S. aureus* being the predominant species observed during flares [5]. The increased proportion of *S. aureus* is accompanied by a reduction in Shannon diversity, an index of microbial diversity that accounts for species richness and evenness [5]. *S. aureus* occupies both the lesional and non-lesional sites, and its abundance is correlated with disease severity [99, 100]. Approximately 70% of patients with AD harbor *S. aureus* in lesional sites, while 39% contain *S. aureus* in non-lesional sites [101]. Murine models investigating the interaction between mast cells and *S. aureus* in skin inflammation have primarily been targeted toward AD.

Nakamura et al. showed that δ-toxin induces allergic skin inflammation in mice via mast cell degranulation (Figure 1) [102]. BALB/c mice epicutaneously sensitized with WT *S. aureus* strain LAC developed significantly greater disease severity and inflammatory cell infiltrate compared to mice treated with δ-toxin-deficient *S. aureus*. The phenotype was abrogated when mast cell-depleted *Kit*$^{W\text{-}sh/W\text{-}sh}$ mice were used, demonstrating that δ-toxin-induced skin inflammation is dependent on mast cells. However, Deng et al. found no significant difference in disease score between WT C57BL/6 mice and *Kit*$^{W\text{-}sh/W\text{-}sh}$ mice epicutaneously sensitized with MRSA [103]. While Nakamura et al. tape-stripped the mice to induce skin barrier disruption prior to the delivery of *S. aureus*, there is no mention of this technique in the paper by Deng et al., which could account for this discrepancy. It is plausible that prior barrier damage is a requisite for either *S. aureus* or *S. aureus*-derived toxins to reach the dermis and activate mast cells. In fact, mutations in the skin structural protein filaggrin are known to augment the penetration of *S. aureus*

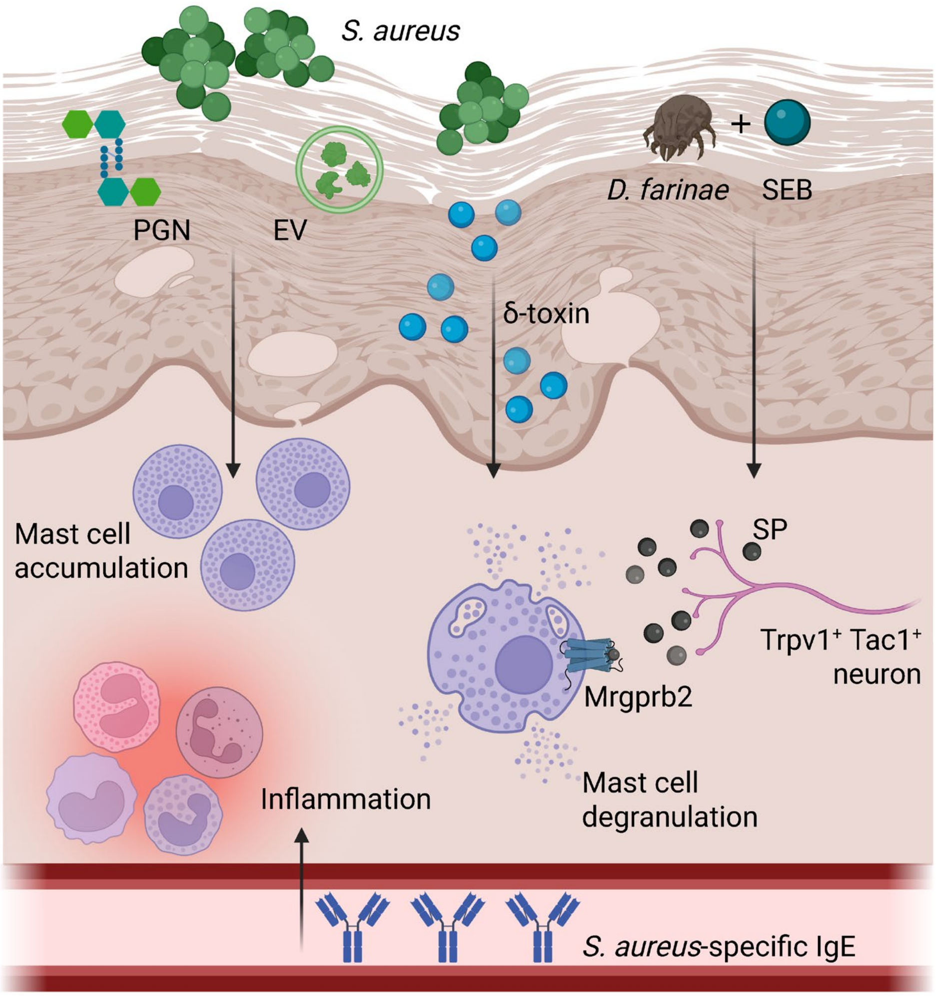

FIGURE 1 | Impact of *S. aureus* epicutaneous sensitization models on mast cell function in vivo. *D. farinae* and Staphylococcal enterotoxin B (SEB) activate Trpv1+ Tac1+ nociceptors to trigger the release of substance P (SP), which promotes mast cell degranulation through Mrgprb2. *S. aureus* extracellular vesicles (EV) and peptidoglycan (PGN) promote mast cell accumulation in the dermis. Finally, δ-toxin released by *S. aureus* promotes mast cell degranulation through an unknown mechanism. These toxins/factors also lead to the development of serum IgE specific to *S. aureus*, marked skin inflammation, and recruitment of inflammatory leukocytes into the dermis. This figure was generated with BioRender.com.

into the dermis of mice [100]. Furthermore, Matsumoto et al. detected *S. aureus* exclusively in the epidermis following epicutaneous sensitization and evidence of mast cell migration to the epidermis during AD is limited [49, 104].

Similar to Gram-negative bacteria, *S. aureus* produces outer membrane-like vesicles called extracellular vesicles (EVs) which harbor proteins, DNA, RNA, and toxins [105, 106]. These EVs contain pathogenic molecules associated with the pathogenesis of AD, including α-hemolysin and cysteine protease [105, 107–109]. In AD, EVs are involved in keratinocyte cell death, necrosis, skin barrier disruption, epidermal hyperplasia, and eosinophilic dermal infiltration [106]. AD patients were shown to harbor *S. aureus*-derived EVs on lesional skin and serum extracellular vesicle-specific IgE [106]. Topical application of *S. aureus* extracellular vesicles to tape-stripped mice induced AD-like skin inflammation characterized by epidermal thickening and the recruitment of inflammatory cells into the dermis (Figure 1) [106, 110]. *S. aureus* EVs resulted in significant mast cell accumulation within the dermis compared to saline-treated mice following both 4 and 8 weeks of exposure [107]. This study demonstrates that mast cells are involved in a chronic phenotype of *S. aureus*-induced AD, in comparison with most studies which only explore a 7-day timepoint. Long-term repeated exposure to *S. aureus* EVs also significantly enhanced serum IgE levels. However, it remains to be determined whether mast cells become sensitized with EV-specific IgE and what role this could play in perpetuating the inflammatory response. Future studies should discern the mechanism through which mast cells increase in the dermis, such as progenitor infiltration or local proliferation.

Several studies have investigated the contribution of mast cells to AD-like skin inflammation triggered by *S. aureus* components either with or without allergen sensitization [111–115]. In one study, Matsui et al. showed that topical application of peptidoglycan (PGN) to barrier-disrupted abdominal skin induced an increase in dermal mast cell numbers, likely through transforming growth factor-β1 release from epidermal keratinocytes [114]. A subsequent study revealed that percutaneous PGN application also increased immature mast cell numbers at a distant site, specifically spleen (Figure 1) [115]. Ando et al. epicutaneously sensitized mice with *Dermatophagoides farinae* extract and the *S. aureus* superantigen staphylococcal enterotoxin B (SEB) to induce a phenotype that resembles human AD [111]. They showed that *Kit*$^{W\text{-}sh/W\text{-}sh}$ mice had significantly reduced disease severity and skin thickening compared to WT mice [111]. Serhan et al. showed that mast cell-mediated skin inflammation in response to *D. farinae* + SEB is dependent on expression of the Mrgprb2 receptor. Using intradermal injections, they demonstrated that *D. farinae* + SEB activated TRPV1^{+} Tac1^{+} peptidergic nociceptors to trigger the release of the neuropeptide and known Mrgprb2 agonist, substance P (Figure 1). *Mrgprb2*$^{+/+}$ mice experienced significantly greater mast cell degranulation, disease severity, and leukocyte recruitment compared to *Mrgprb2*$^{mut/mut}$ mice [112]. Overall, this study demonstrates that epicutaneous treatment of mice with *D. farinae* and SEB exacerbates skin inflammation through neurogenic mast cell activation of the Mrgprb2 receptor. Despite this interesting finding, this study is not without limitations. Although *D. farinae* + SEB induced more severe skin inflammation compared to either component alone, there was no significant difference in substance P production in response to *D. farinae* alone or *D. farinae* + SEB. We therefore cannot rule out the possibility that SEB exacerbates skin inflammation through a mechanism that is independent of the Mrgprb2 receptor. Furthermore, neither of these studies treated mice with live *S. aureus* + *D. farinae* which would more closely resemble the pathogenesis of AD.

A major limitation of *S. aureus* epicutaneous sensitization models is their overreliance on a single aspect of the disease. AD is a complex and multifactorial disease involving genetics, epidermal barrier defects, microbial dysbiosis, IgE sensitization, a T helper 2 (Th_2) skewed immune response, and environmental factors [5, 116–119]. While focusing solely on *S. aureus* is necessary to isolate its direct impact on mast cell function, these models are likely to overconflate the role this interaction plays in clinical settings. Furthermore, these models are self-resolving and use Tegaderm to temporarily occlude the site of infection which are features not observed in human AD [103].

7 | Mast Cells in *S. aureus* Intradermal Infections

As previously mentioned, *S. aureus* can establish deeper SSTIs that reach the dermis. To mimic these infections, Liu et al. infected WT and *Kit*$^{W\text{-}sh/W\text{-}sh}$ mice intradermally with an *S. aureus* strain from a clinical isolate. *Kit*$^{W\text{-}sh/W\text{-}sh}$ mice developed significantly greater lesion sizes compared to WT mice, which were indistinguishable by day 14. On day 3 post-infection, *Kit*$^{W\text{-}sh/W\text{-}sh}$ mice had a 10-fold higher bacterial load and significantly reduced neutrophil recruitment in comparison to WT mice. *Kit*$^{W\text{-}sh/W\text{-}sh}$ mice also had reduced production of the pro-inflammatory cytokines, IL-6, TNF, and cathelicidin-related antimicrobial peptide (CRAMP) on day 3 (Figure 2) [120]. It was further shown using co-immunostaining that mast cells are a major source of TNF in infected skin, which is known to enhance Th17-dependent neutrophil infiltration in other pathologies [120–122]. Treatment with recombinant TNF largely restored the phenotype in *Kit*$^{W\text{-}sh/W\text{-}sh}$ mice to

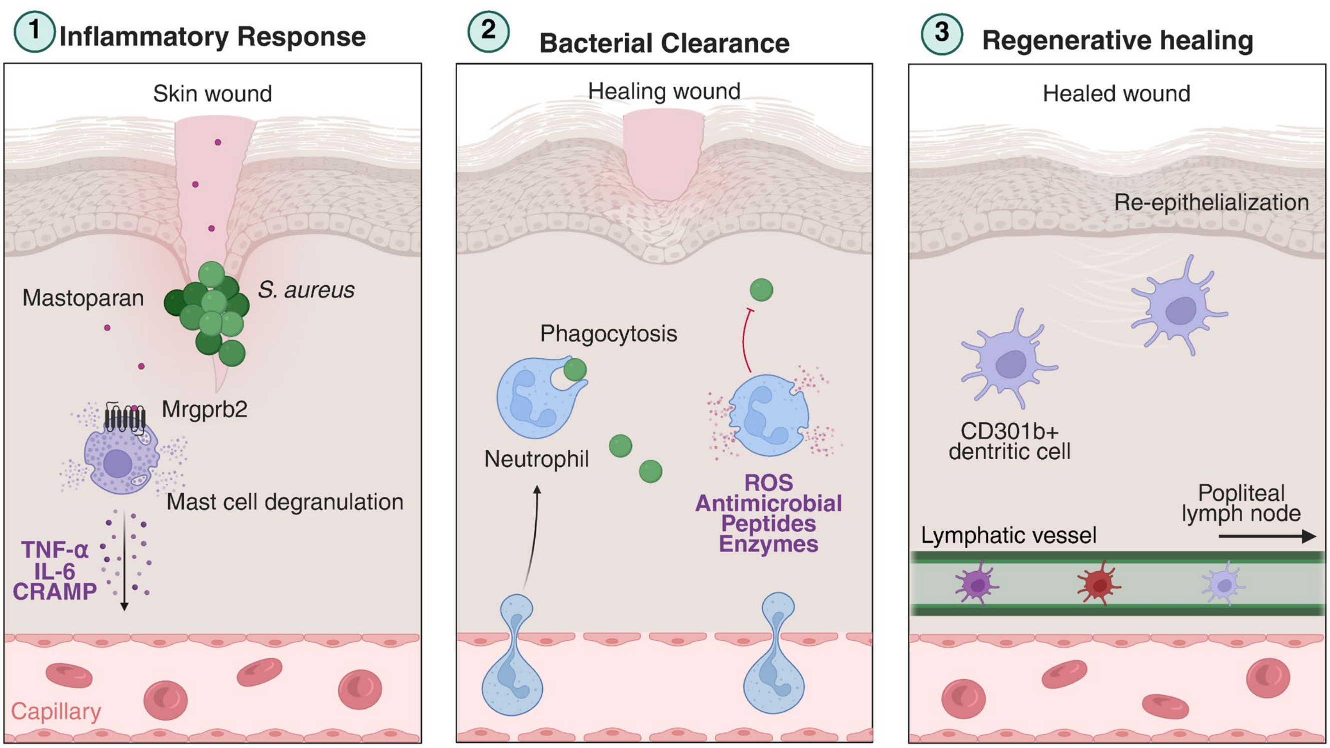

FIGURE 2 | Mastoparan enhances the endogenous immune response to intradermal *S. aureus* infection. Mastoparan activates Mrgprb2 within mast cells, resulting in mast cell degranulation (1). Mast cell mediators act on the vasculature to increase vascular permeability and facilitate neutrophil diapedesis into infected skin. Neutrophils employ a variety of antibacterial mechanisms, such as phagocytosis and extracellular trap formation to clear *S. aureus* (2). Mast cells promote regenerative wound healing via CD301b^{+} dendritic cells and enhance the trafficking of dendritic cell populations into the popliteal lymph node (3). This figure was generated with BioRender.com.

that of WT mice [120]. Although this is a promising finding, it cannot be said for certain that a causal relationship exists between mast cell-derived TNF and enhanced lesion closure or decreased CFU. To provide a causal conclusion that mast cell-derived TNF is mediating this phenotype, the authors could have reconstituted BMMC from TNF-knockout mice into $Kit^{W\text{-}sh/W\text{-}sh}$ mice. Based on the difference in TNF production between WT mice and $Kit^{W\text{-}sh/W\text{-}sh}$ mice in this study, mast cells produce approximately 200 pg/mg of infected tissue. The authors chose to give a dose of 200 ng, but it is unclear if this dose reflects the amount of TNF that mast cells would be producing physiologically. Therefore, this study may be overestimating the role that TNF plays in comparison with other mast cell-derived mediators.

These findings were corroborated by Arifuzzaman et al. using mast cell-deficient *Mcpt-Cre*$^{+}$ *iDTR*$^{+}$ and *Mcpt5-Cre*$^{-}$ *iDTR*$^{+}$ WT mice. Interestingly, neutrophil depletion abrogated the differences in lesion size 24 h after infection, demonstrating that mast cells primarily impact intradermal *S. aureus* infection through neutrophil recruitment. They also showed that mastoparan, an agonist of receptors MRGPRX2/b2, can be used as a pharmacological treatment for intradermal *S. aureus* infections. Treating lesions topically with mastoparan accelerated bacterial clearance and wound healing via neutrophil recruitment (Figure 2). Furthermore, it was shown that mast cells recruit dermal CD301b^{+} dendritic cells, which promote regenerative healing through reepithelialization (Figure 2). This process was enhanced by treatment with mastoparan, resulting in significantly reduced scar formation. Intriguingly, in a mast cell-specific manner, mastoparan recruited migratory dendritic cells into the popliteal lymph node and enhanced the production of *S. aureus*-specific IgG, conferring protection against subsequent infection [123]. Despite the robustness of this study, one limitation was the lack of Mrgprb2-deficient mice in the mastoparan-treated group. We therefore cannot exclude the possibility that mastoparan has direct effects on other immune cells, such as neutrophils and dendritic cells, which are of relevance to this model. An important point to consider is the difference in the concentration of mastoparan used across experimental contexts in the study. The authors describe using a dose of 10 μg/μl to induce mast cell degranulation in the skin, which corresponds to approximately 6761.78 μM [123]. This is substantially higher than the EC_{50} for mastoparan (24 μM) observed in Mrgprb2-expressing HEK293 cells [31]. While the exact concentration of mastoparan applied topically to infected lesions is not clearly stated, if the same 10 μg/μl dose was used, it would exceed the minimum inhibitory concentration for *S. aureus*, which ranges from 10.8 to 21.6 μM. This raises the possibility that the observed effects on lesion closure may, at least in part, be influenced by the antimicrobial properties of mastoparan, potentially acting as a confounding factor in the interpretation of the results. They do, however, show that a derivative of mastoparan, MP-6I, which has antibacterial activity but is significantly reduced in its ability to activate mast cells, is no more effective than vehicle in reducing the size of infected lesions [123].

Amponnawarat et al. engineered a synthetic analog of the antimicrobial peptide DJK5 called $[\text{Pro}]^7$-DJK5. While both DJK5 and $[\text{Pro}]^7$-DJK5 were found to be agonists of MRGPRX2/b2, $[\text{Pro}]^7$-DJK5 lost its direct antimicrobial activity, enabling selective study of its immunomodulatory capacity. Upon intradermal infection with *S. aureus*, $[\text{Pro}]^7$-DJK5 recruited neutrophils and monocytes in WT mice but not in Mrgprb2-deficient mice. As well, $[\text{Pro}]^7$-DJK5 significantly reduced skin bacterial load in WT mice compared to vehicle-treated mice, but not in Mrgprb2-deficient mice. However, no difference was observed in the bacterial load between WT mice and Mrgprb2-deficient mice treated with $[\text{Pro}]^7$-DJK5 [124]. It therefore cannot be said for certain whether $[\text{Pro}]^7$-DJK5 is able to mediate bacterial clearance through Mrgprb2. One would expect the bacterial load in $[\text{Pro}]^7$-DJK5-treated Mrgprb2-deficient mice to mirror that of vehicle-treated WT mice. In a related study, Zhang et al. demonstrated that long-term ablation of Langerhans cells and nonpeptidergic MrgprD-expressing skin neurons significantly reduced lesion sizes and CFU following intradermal *S. aureus* injection. This mechanism was believed to be mediated by increased expression of Mrgprb2 as a consequence of the ablation [52]. Overexpression of dermal Mrgprb2 may enhance activation by endogenous ligands, providing further proof of this receptor's therapeutic potential.

In addition to treatment with live *S. aureus*, there has been a growing interest in studies investigating the injection of *S. aureus* components directly into the dermis. In response to intradermal injection of α-hemolysin, mast cells promoted a chronic anti-inflammatory phenotype [65]. Contrarily, mast cells were shown to mediate skin inflammation after intradermal injection of streptolysin O, a toxin produced by the common skin pathogen *Streptococcus pyogenes*. These results demonstrate that mast cells can exhibit opposing responses depending on the type of bacterial toxin [65, 125]. Intradermal injection of PGN was shown to induce mast cell degranulation, vasodilation, and neutrophil recruitment in a TLR2-dependent manner [126]. Mast cells have also been shown to mediate the mobilization of dendritic cell subpopulations into the draining lymph node following PGN injection [127]. Although these studies provide greater insight into the interaction between mast cells and *S. aureus* in the skin, they are likely to oversimplify the interactions that would happen in the context of infection. Furthermore, it is unclear whether mast cells are active contributors to host defense in these models through promoting bacterial clearance or tissue regeneration, for instance.

8 | Mast Cells in *S. aureus* Subcutaneous Infections

Staphylococcus aureus is a causative agent in a variety of skin infections that penetrate into the subcutaneous layer of the skin, such as furuncles (boils), carbuncles, abscesses, and cellulitis [128]. To date, there have been very few studies examining the interaction between mast cells and *S. aureus* during subcutaneous infections. In an intra-abdominal abscess model, subcutaneous infection with *S. aureus* resulted in mast cell distribution around the abscess and degranulation [129]. Across the full 10-day study period, mice infected with *S. aureus* had significantly greater mast cell density in the skin compared to saline-treated control mice, which correlated positively with expression of the mast cell granule protease tryptase [129]. In vivo subcutaneous infection resulted in the internalization of *S. aureus* within mast cells [60]. In accordance, mast cells were shown to upregulate the expression of β1 integrin (CD29) following subcutaneous infection with WT *S. aureus* but not *S. aureus* deficient in α-hemolysin (*hla* mutant) [62]. This is in agreement with in vitro data demonstrating that mast cells internalize *S. aureus* by upregulating β1 integrin in response to α-hemolysin [60, 62]. However, a comprehensive study investigating *S. aureus* internalization within mast cells treated with WT *S. aureus* versus hla-mutant *S. aureus* is still required to confirm this. An important drawback of these studies is the inability to discern whether mast cells play a direct role in defense against *S. aureus*. It is possible that mast cell activation is a consequence of abscess formation rather than a causal contributor to host defense [88].

9 | Mast Cells in Acquired Host Defense Against Cutaneous *S. aureus*

Many of the aforementioned studies have overlooked the potential for a mast cell-dependent acquired host defense axis in response to *S. aureus*. Starkl et al. epicutaneously sensitized the backs of C57BL/6 mice for 7 days followed by a subsequent ear skin and soft tissue (ESST) infection on day 28 to induce a secondary immune response. Previously sensitized mice experienced a reduction in ear and lymph node bacterial load and tissue loss relative to mock-infected mice. Remarkably, this protection was also present at a distant infection site, specifically the lungs. Mice lacking functional IgE (*Igh-7*$^{-/-}$), FcεRIα (*Fcer1a*$^{-/-}$), or mast cells (*Mcpt5-cre*) failed to mount this protective immunity, suggesting the existence of an IgE-FcεRIα-mast cell axis [130]. Future experiments would benefit from mast cell-specific FcεRIα knockout mice, such as *FcεRIα*$^{fl/fl}$ mice crossed with *Mcpt5-Cre* transgenic mice, to substantiate this theory. Furthermore, the authors could have repeated the experiment with more time between the initial sensitization and the follow-up infection, such as months or even a year, to determine if the protection is long lasting. Another thing to consider is the epicutaneous model used for the initial infection, which is superficial and commonly used as a model for AD. It has been reported in the literature that individuals with AD have a higher prevalence of severe skin, multiorgan, and systemic infection, which may undermine the clinical relevance of these results [131, 132].

10 | Future Directions

10.1 | Mast Cells and Bacteremia/Sepsis

A crucial area that should be explored in future studies is whether skin mast cells impact susceptibility to bacteremia or sepsis instigated by *S. aureus*. Studies exploring the mast cell-mediated response in bacteremia or sepsis have been contradictory and may be influenced either by the type of bacterial pathogen or site of infection. For instance, in response to subcutaneous group A streptococcal infection, mast cells were shown to limit bacterial dissemination into the blood and spleen [58]. In the peritoneal cavity, mast cells aggravated sepsis by inhibiting macrophage phagocytosis of *Escherichia coli* through TLR4-dependent IL-4 production [133]. *S. aureus* bacteremia has a mortality rate of 20%–30%, with a striking increase to 20%–50% for MRSA infections [134–137]. It is therefore of utmost importance to contain *S. aureus* before it penetrates the blood from the subcutaneous tissue.

10.2 | Mast Cells in Infected Excisional Wounds

Infected excisional wounds pose a serious health risk, especially for individuals with conditions that delay wound healing, such as diabetes [138], obesity [139], and stress [140]. Although mast cells have been implicated in host defense at barrier tissues, their role in infected excisional wounds remains incompletely defined and may depend on the infecting organism, the inflammatory context, and the mode of mast cell activation. In *S. aureus*-infected skin wounds, recent work from Guth et al. demonstrated that mast cells are activated and accumulate within infected wounds, and that engagement of the mast cell Mrgprb2 pathway by the neuroendocrine peptide catestatin promotes bacterial clearance. This response was associated with reduced MRSA burden, altered inflammatory cytokine production and leukocyte infiltration, and increased expression of the antimicrobial peptide *Defb14*, suggesting that mast cells can shape both antimicrobial and inflammatory responses during cutaneous infection [141]. Consistent with this, previous work showed that mast cells control *S. aureus* bacterial load in ex vivo-infected skin explants. In a mouse model using both mast cell-deficient, *Kit*$^{W/W\text{-}v}$ and *Cpa3-Cre; Mcl-1*$^{fl/fl}$ mice, mast cells accelerated the healing and bacterial clearance of skin wounds infected by *P. aeruginosa*. This mechanism was dependent on mast cell-derived IL-6 which mediated the release of antimicrobial peptides from keratinocytes. Intriguingly, the improved antibacterial response observed in WT mice compared to *Kit*$^{W/W\text{-}v}$ mice was independent of neutrophils [142]. The discovery that mast cells promote the clearance of *S. aureus* in ex vivo-infected skin biopsies suggests that neutrophils are also dispensable in this context. Nonetheless, future studies should examine whether mast cells are required for proper healing and bacterial elimination in *S. aureus* wound infections in vivo, and whether this mechanism is dependent on IL-6 release and crosstalk with keratinocytes.

10.3 | Crosstalk Between Mast Cells, *S. aureus* and Sensory Neurons

While there is evidence that Mrgprb2 is activated by substance P released from nociceptive sensory neurons in response to SEB and *D. farinae* [112], there is little known about the interaction between live *S. aureus*, mast cells, and neurons in the skin. One study showed that WT mice reinfected with *S. aureus* following previous sensitization experienced significantly greater mast cell degranulation, dye extravasation, ear thickening, and TNF expression compared to mice lacking the pruriceptive receptor Mrgpra3 or physically restricted from scratching [143]. However, the mechanism through which this occurs remains elusive. For instance, does *S. aureus*-specific IgE activate mast cells, which in turn activate Mrgpra3-expressing neurons that perpetuate mast cell degranulation? Or does *S. aureus* activate Mrgpra3-expressing neurons directly, thereby enhancing IgE-mediated mast cell degranulation? Future studies should also investigate the interaction between mast cells, *S. aureus*, and sensory neurons in deeper skin infections, as they can be a significant source of pain [144].

11 | Conclusion

In this review, we have discussed the mast cell-mediated immune response against *S. aureus* in three skin infection models discussed in the literature: epicutaneous sensitization, intradermal infection, and subcutaneous infection. In 2019, *S. aureus* SSTIs had an all-cause age-standardized mortality rate of 0.5 per 100 000 population, underscoring the need for improved treatment options [145]. This concern is confounded by the antibiotic-resistant nature of *S. aureus*, with deaths attributable to antimicrobial resistance increasing from 103 000 in 1990 to a staggering 196 000 in 2021 [146]. Research on the mast cell *S. aureus* crosstalk in skin infection is still in its preliminary stages, and it is our hope to bring attention to this rapidly expanding field of research.

Author Contributions

Hannah Dychtenberg: conceptualization, writing – original draft, writing – review and editing. **Carly Kadonoff:** writing – review and editing. **Bailey Lehnert:** writing – review and editing. **Priyanka Pundir:** conceptualization, funding acquisition, writing – review and editing, project administration, supervision, writing – original draft, resources.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The authors have nothing to report.

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